

MultiFPT: Towards Multi-Attribute Fairness in Pre-Trained Graph Neural Networks via Prompt Tuning

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Abstract

Pre-trained Graph Neural Networks (GNNs) have demonstrated remarkable performance in graph mining tasks, yet they often amplify societal biases against protected demographic groups. Existing fairness-aware approaches primarily address discrimination based on a single sensitive attribute (e.g., gender or race), overlooking real-world scenarios where individuals possess multiple overlapping demographic characteristics, leading to unfair treatment of underrepresented subgroups. Moreover, incorporating extra fairness constraints into pre-trained GNNs usually requires full model retraining, which is computationally expensive and often impractical. To address these limitations, we propose a novel Multi-attribute Fairness-aware Prompt Tuning framework named MultiFPT¹. Our approach operates in two key stages: in the graph prompt learning stage, MultiFPT injects fairness-aware structural and feature prompts into pre-trained GNN inputs; in the adapter tuning stage, a lightweight adapter regularized by the Hilbert–Schmidt Independence Criterion (HSIC) enforces statistical independence between node representations and multiple sensitive attributes. Experiments on real-world datasets demonstrate that MultiFPT significantly improves multi-attribute fairness, reducing bias by approximately 30% on average in node classification while maintaining competitive predictive performance compared to state-of-the-art baselines.

CCS Concepts

• Information systems → Data mining; • Applied computing
→ Law, social and behavioral sciences.

*Demin Gao is the corresponding author.

¹The source code is available at <https://github.com/mengcao327/MultiFPT>.



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Keywords

Graph Neural Networks; Fairness; Graph Prompting; Network Representation Learning

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1 INTRODUCTION

Graph Neural Networks (GNNs) have emerged as a powerful tool for graph representation learning, achieving remarkable success in analyzing graph-structured data, such as social networks [47] and webpage networks [44]. To reduce the reliance on large amounts of labeled data, various pre-trained graph models (PGMs) have been developed. These models capture universal graph knowledge from unlabeled data through self-supervised learning, such as GraphCL [46] and GPT-GNN [19]. PGMs can be efficiently adapted to downstream tasks via fine-tuning, achieving strong predictive performance in various domains, such as recommender systems [30] and financial risk assessment [29].

However, recent studies [1, 6, 48] show that GNNs and PGMs are susceptible to biases in graph data. They often inherit and even amplify social biases during representation learning, leading to discriminatory predictions against certain demographic groups. This raises serious ethical concerns, especially in high-stakes applications such as credit scoring and criminal justice. While a growing body of work aims to mitigate such biases, most focus on fairness with respect to a *single* sensitive attribute (e.g., gender or race). Nevertheless, real-world individuals usually possess multiple overlapping demographic characteristics (e.g., a *young black woman*). A critical question thus arises: *Can models optimized for single-attribute fairness effectively mitigate bias on other attributes?*

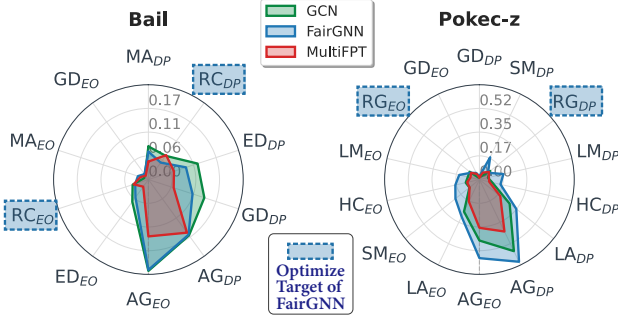


Figure 1: Multi-attribute fairness evaluation on Bail [21] and Pokec-z [36] with Δ_{DP} ($\downarrow$) and Δ_{EO} ($\downarrow$). Sensitive attributes are abbreviated (e.g., RC: Race, RG: Region, etc.). It shows that FairGNN [6] which focuses on a single sensitive attribute, may exacerbate bias on unconsidered attributes. In contrast, MultiFPT improves fairness across all sensitive attributes.

To answer this question, we evaluate the multi-attribute fairness of GCN [24] and FairGNN [6] on the Bail [21] and Pokec-z [36] datasets. As shown in Figure 1, while FairGNN reduces bias for the target attributes *Race* (RC) and *Region* (RG), it inadvertently *amplifies* bias on other unconsidered attributes compared to the vanilla GCN. This observation highlights a fundamental challenge: *How can we achieve fairness across multiple sensitive attributes simultaneously?* Although many fairness-aware methods have been proposed, including graph modification [33, 35], adversarial debiasing [6], and fair message-passing schemes [25], they face three major limitations in the context of multi-attribute fairness. *First*, methods designed for a single attribute often fail to generalize and may harm fairness on other attributes [4]. *Second*, directly extending the fairness objectives to multiple sensitive attributes can lead to the *fairness gerrymandering* [23] issue, where certain intersectional subgroups (e.g., “Black Women”) are unfairly treated even when predictions appear fair for each attribute individually. *Third*, for PGMs that are usually optimized for predictive utility, introducing new fairness constraints requires full retraining, which is computationally expensive and impractical for real-world deployment.

To address these issues, we propose **Multi-attribute Fairness-aware graph Prompt Tuning (MultiFPT)**, a novel framework designed to endow PGMs with multi-attribute fairness without retraining. The core idea is to inject lightweight and learnable prompts into the inputs of PGMs, guiding the model to produce fairer outcomes through prompt tuning. Specifically, MultiFPT operates in two stages: (1) in the *graph prompt learning* stage, a *distance-guided structure prompting* module mitigates topological bias by encouraging heterophilic connections among demographically distant nodes; and a *contrastive mixed feature prompting* module adjusts node features to reduce feature bias and generate counterfactual augmented samples. (2) In the *adapter tuning* stage, a *multi-fairness-aware adapter training* module uses the Hilbert-Schmidt Independence Criterion (HSIC) to disentangle the correlations between the learned network representations and all sensitive attributes.

Our main contributions are summarized as follows:

- We first investigate the fairness of pre-trained GNNs with multiple sensitive attributes. Our findings reveal that existing fair

GNNs fail to address multi-attribute bias, leading to biased outcomes in downstream tasks.

- We propose MultiFPT, a novel graph prompt tuning framework that achieves multi-attribute fairness by optimizing lightweight prompts and an HSIC-based adapter, without retraining the backbone GNN model.
- Comprehensive experiments on real-world datasets demonstrate that MultiFPT outperforms state-of-the-art baselines in improving multi-attribute fairness while maintaining utility.

2 RELATED WORK

2.1 Graph Neural Networks

Graph Neural Networks (GNNs) have been well-studied for graph data mining. Existing methods primarily follow two paradigms: *spectral-based* and *spatial-based* approaches. Spectral-based GNNs [3] perform graph convolutions in the spectral domain, including ChebNet [7] and Graph Convolutional Networks (GCN) [24]. In contrast, spatial-based methods perform convolutions directly on graph structures via message-passing, including GraphSAGE [13] and Graph Attention Networks (GAT) [38]. Most GNNs rely on supervised learning, which demands substantial labeled data. This limitation motivates *self-supervised learning* (SSL) methods that leverage unlabeled graph data through pretext tasks. For instance, DGI [39] maximizes mutual information between local node representations and a global graph summary, while GraphMAE [17] employs masked feature reconstruction. These SSL techniques facilitate transferable representation learning, paving the way for large-scale graph pre-training.

2.2 Pre-trained Graph Models

Building upon the success of self-supervised learning, pre-trained graph models (PGMs) learn universal graph representations transferable to various downstream tasks. Existing PGMs primarily fall into two categories: *contrastive* and *generative* methods. *Contrastive methods* learn representations by contrasting positive and negative node pairs, such as GraphCL [46] and JOAO [45]. *Generative methods* learn by predicting masked or corrupted graph components. For example, GPT-GNN [19] employs self-supervised attributed graph generation to capture structural and semantic properties, while Molca [31] focuses on molecule-text retrieval through cross-modal projection. To adapt PGMs to downstream tasks, two primary paradigms exist: *fine-tuning* and *prompt tuning*. *Fine-tuning* updates most or all PGM parameters [18], but it is computationally expensive and may cause catastrophic forgetting. Inspired by natural language processing (NLP), *prompt tuning* has emerged as a parameter-efficient alternative. Methods like GraphPrompt [32] and GPF [11] learn lightweight prompts to adjust inputs without retraining the PGMs. However, these methods prioritize task accuracy and often overlook social biases in the original models, potentially leading to unfair predictions. This gap motivates the development of fairness-aware adaptation approaches.

2.3 Fairness in Graph Neural Networks

Extensive research demonstrates that GNNs can amplify social biases, resulting in discriminatory predictions based on sensitive attributes like gender or race [1, 2, 6]. Existing fairness-aware

methods include: *pre-processing* techniques that debias graph data through edge editing or feature manipulation [5, 35]; *in-processing* methods incorporating fairness constraints during training [6, 15, 22, 28]; and *post-processing* approaches adjusting predictions after training [26, 34, 41]. However, these methods face two critical limitations. First, they require task-specific end-to-end training, making them incompatible with PGMs. Second, they predominantly address single-attribute fairness, compromising fairness on other sensitive attributes [23]. Recent multi-attribute fairness-aware approaches like FairMILE [16] and FairINV [49] still require complete retraining for new fairness constraints, limiting their practical applicability.

Our work addresses these gaps by proposing a plug-and-play module compatible with any PGM. It promotes multi-attribute fairness through prompt tuning within a unified framework, offering an efficient and adaptable solution to fair graph learning.

3 PRELIMINARY

3.1 Notations

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote a graph with node set $\mathcal{V} = \{v_1, \dots, v_N\}$ and edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. The total number of nodes is $N = |\mathcal{V}|$. Each node $v_i \in \mathcal{V}$ is associated with a feature vector $\mathbf{x}_i \in \mathbb{R}^d$, and the feature matrix for all nodes is denoted as $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$. The graph structure is described by an adjacency matrix $\mathbf{A} \in \{0, 1\}^{N \times N}$, where $A_{ij} = 1$ if $(v_i, v_j) \in \mathcal{E}$, and $A_{ij} = 0$ otherwise. Each node v_i possesses a sensitive attribute vector $\mathbf{s}_i = \{s_i^1, s_i^2, \dots, s_i^K\}$, representing its membership in K overlapping demographic groups (e.g., gender, race). For simplicity, we assume all sensitive attributes are binary², i.e., $s_i^k \in \{0, 1\}, \forall k \in [1, K]$. The sensitive attribute matrix for the entire graph is denoted as $\mathbf{S} = \{\mathbf{s}_1, \dots, \mathbf{s}_N\}$. In this work, we focus on the node classification task, where a subset of nodes $\mathcal{V}_L \subseteq \mathcal{V}$ is labeled. Each node $v_i \in \mathcal{V}_L$ has a ground-truth label y_i from the label set $\mathcal{Y} = \{y_1, \dots, y_{|\mathcal{V}_L|}\}$.

3.2 Graph Prompt Tuning

We consider a pre-trained GNN model f_θ , which has been optimized through a self-supervised objective $\mathcal{L}_{ssl}(f_\theta(\mathbf{X}, \mathbf{A}))$. During the prompt tuning process, the parameters θ of f_θ are frozen. A prompting function Φ transforms the original input graph $\mathcal{G}$ into a prompted graph $\mathcal{G}^*$, while a lightweight adapter Ψ maps the GNN's output representations to task-specific predictions. The final prediction for node v_i is given by $\hat{y}_i = \arg \max(\mathbf{z}_i)$, where $\mathbf{z}_i$ is the adapted output. Both the prompt Φ and the adapter Ψ are jointly optimized for specific downstream tasks.

3.3 Fairness Measurement

We adopt the notion of *group fairness*, which requires equitable treatment across different demographic groups [6]. For binary node classification ($y \in \{0, 1\}$), we employ two fairness metrics:

Definition 3.1 (Demographic Parity [10]). Demographic parity requires that the positive prediction rate should equal across groups:

$$P(\hat{y} = 1 | s = 0) = P(\hat{y} = 1 | s = 1). \quad (1)$$

²Our method naturally supports multi-class sensitive attributes within the same computational framework.

Definition 3.2 (Equal Opportunity [14]). Equal opportunity requires equality of true positive rates across groups:

$$P(\hat{y} = 1 | y = 1, s = 0) = P(\hat{y} = 1 | y = 1, s = 1). \quad (2)$$

Following existing work [1, 6], we quantify fairness for the k -th sensitive attribute $\mathbf{s}^k$ with the following disparity metrics:

$$\Delta_{DP}^k = |P(\hat{y} = 1 | \mathbf{s}^k = 0) - P(\hat{y} = 1 | \mathbf{s}^k = 1)|, \quad (3)$$

$$\Delta_{EO}^k = |P(\hat{y} = 1 | y = 1, \mathbf{s}^k = 0) - P(\hat{y} = 1 | y = 1, \mathbf{s}^k = 1)|. \quad (4)$$

Most importantly, to obtain an overall assessment of fairness across all K sensitive attributes, we compute the average disparities:

$$\bar{\Delta}_{DP} = \frac{1}{K} \sum_{k=1}^K \Delta_{DP}^k, \quad \bar{\Delta}_{EO} = \frac{1}{K} \sum_{k=1}^K \Delta_{EO}^k. \quad (5)$$

Smaller values of $\bar{\Delta}_{DP}$ and $\bar{\Delta}_{EO}$ indicate fairer results.

3.4 Problem Formulation

Given a partially labeled graph $\mathcal{G}$ and sensitive attributes $\mathbf{S} \in \mathbb{R}^{N \times K}$, our objective is to adapt a pre-trained GNN f_θ to downstream node classification tasks without modifying its parameters, while ensuring fairness across all K sensitive attributes. Formally, we aim to learn a prompting function Φ and an adapter Ψ that minimize the overall fairness disparities $\bar{\Delta}_{DP}$ and $\bar{\Delta}_{EO}$ (as defined in Eq. (5)) while maintaining satisfying predictive performance.

4 THE PROPOSED FRAMEWORK

In this section, we propose MultiFPT, a novel graph prompt tuning framework that enhances multi-attribute fairness in PGMs for downstream tasks. As illustrated in Figure 2, the framework consists of three core components: (a) a *distance-guided structure prompting* module that strategically modifies the network structure to enable fairer information propagation; (b) a *contrastive mixed feature prompting* module that mitigates feature-level bias through augmented prompt mix-up techniques; and (c) a *multi-fairness-aware adapter* that ensures statistical independence of embeddings and all sensitive attributes. Each component is described below.

4.1 Distance-Guided Structure Prompting

Structural bias is a key source of unfairness in graph data. Recent works have demonstrated that high *homophily* can cause GNN message-passing to amplify societal disparities [8, 9, 20]. To mitigate structural bias across multiple sensitive attributes, we propose a distance-guided structure prompting method, which enhances *heterophily* by strategically modifying edges based on sensitive attribute distances.

Specifically, for a node pair (v_i, v_j) with sensitive attribute vectors $\mathbf{s}_i, \mathbf{s}_j \in \mathbb{R}^K$, we first compute their Hamming distance d_{ij} in the sensitive attribute space:

$$d_{ij} = \sum_{k=1}^K |s_i^k - s_j^k|. \quad (6)$$

This distance quantifies demographic dissimilarity between the two nodes. To promote connections across different demographic

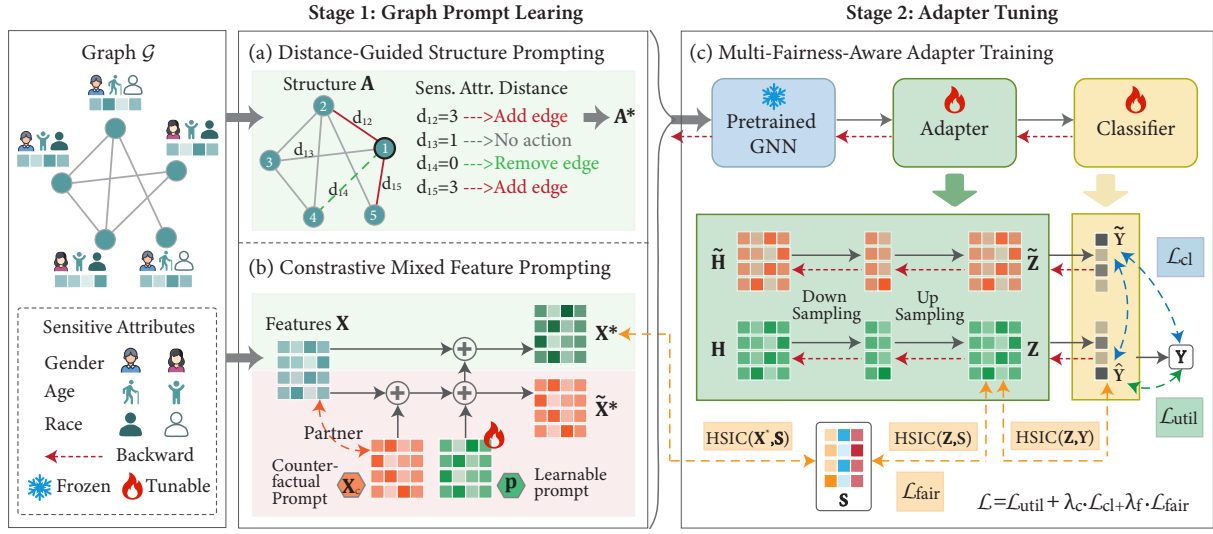


Figure 2: Overview of MultiFPT. The framework operates through two stages: (1) in the *graph prompt learning* stage, structure and feature prompts are generated as inputs to the pre-trained GNNs; (2) in the *adapter tuning* stage, both the output embeddings and feature prompts are jointly optimized to enhance multi-attribute fairness.

groups, we define the edge addition probability as a scaled sigmoid function of d_{ij} :

$$\mathcal{P}(d_{ij}) = \frac{1}{1 + e^{-\eta(2d_{ij}-1)}}, \quad (7)$$

where η is a scaling factor that controls the sensitivity of the probability curve. For non-adjacent node pairs, a larger Hamming distance d_{ij} yields a higher probability of edge addition.

Conversely, for existing edges, the removal probability is defined as $1 - \mathcal{P}(d_{ij})$. In this way, edges connecting demographically similar nodes (with small d_{ij}) are more likely to be removed. The transformation from the original adjacency matrix A to the prompted adjacency matrix A^* is formalized as:

$$A_{ij}^* = \begin{cases} 0 & \text{if } A_{ij} = 1 \wedge u < 1 - \mathcal{P}(d_{ij}), \\ 1 & \text{if } A_{ij} = 0 \wedge u < \mathcal{P}(d_{ij}), \\ A_{ij} & \text{otherwise,} \end{cases} \quad (8)$$

where $u \sim \text{Uniform}(0, 1)$. This approach flexibly promotes heterophily across multiple sensitive attributes. In practice, we limit the total number of edge modifications to no more than 50% of the original edges to avoid excessive distortion of the graph.

As illustrated in Figure 2 (a), a large Hamming distance (e.g., $d_{12} = 3$) between node 1 (*Female, Young, Black*) and node 2 (*Male, Old, White*) promotes edge addition, while a small Hamming distance (e.g., d_{14}) increases edge removal likelihood. This strategic modification fosters heterophilic connections, facilitating fairer information propagation across demographic groups.

While structure prompting addresses topological bias in graph data, node features constitute another critical source of unfairness [8]. To tackle this issue, we introduce a feature prompting method in the next subsection.

4.2 Contrastive Mixed Feature Prompting

Feature prompting is an effective strategy in graph prompt learning, where node features are augmented with learnable vectors to adapt PGMs to downstream tasks with minimal parameter updates [11]. However, existing methods often overlook inherent feature biases, which can be amplified during prompting and lead to unfair predictions. While recent efforts [27, 48] have attempted to mitigate bias for a single sensitive attribute, they generally fall short in multi-attribute scenarios.

To tackle this limitation, we propose a contrastive mixed feature prompting method, including two components: (1) a *Learnable Feature Prompting* module that preserves pre-trained knowledge while adapting to downstream tasks, and (2) a *Counterfactual Prompts Augmentation* module that introduces counterfactual feature variations to neutralize inherent feature biases.

4.2.1 Learnable Feature Prompting. To enhance feature adaptability for downstream tasks, we incorporate learnable feature prompts inspired by recent graph prompting techniques [11]. These prompts enable parameter-efficient fine-tuning through a lightweight transformation $X^* = \mathcal{T}(X; \Theta_p)$, where Θ_p denote the parameters of learned prompts, and $\Theta_p \ll \Theta_{GNN}$.

Specifically, we implement the transformation $\mathcal{T}$ using a set of Q basis vectors $P^b = \{p_1^b, \dots, p_Q^b\}$. For each node v_i , its prompt vector is computed as an attention-weighted combination:

$$p_i = \sum_{q=1}^Q \alpha_{i,q} p_q^b, \quad (9)$$

where the attention weights $\alpha_{i,q}$ measure the relevance between node feature x_i and basis vector p_q^b :

$$\alpha_{i,q} = \frac{\exp(a_q^T x_i)}{\sum_{l=1}^Q \exp(a_l^T x_i)}, \quad (10)$$

where $\mathbf{a}_q \in \mathbb{R}^d$ is the learnable parameter. The final prompted feature for node v_i is then obtained by:

$$\mathbf{x}_i^* = \mathbf{x}_i + \mathbf{p}_i. \quad (11)$$

The prompted feature matrix $\mathbf{X}^* = \{\mathbf{x}_1^*, \dots, \mathbf{x}_N^*\} \in \mathbb{R}^{N \times d}$ serves as input to the pre-trained GNN for downstream task learning.

4.2.2 Counterfactual Prompts Augmentation. To enhance multi-attribute fairness and ensure robustness of the learned prompts, we introduce a feature mix-up strategy inspired by counterfactual augmentation in fair graph learning [1]. This approach enforces output consistency between original and counterfactually perturbed inputs, compelling the model to learn invariant features independent of sensitive information.

Formally, for each node v_i with sensitive attribute vector $\mathbf{s}_i = \{s_i^1, \dots, s_i^K\}$, we identify a counterfactual partner v_c from the candidate set $\mathcal{V}_c$ with maximum Hamming distance between their respective attribute vectors:

$$v_c = \arg \max_{v_j \in \mathcal{V}_c} \sum_{k=1}^K |s_i^k - s_j^k|. \quad (12)$$

When multiple nodes achieve the maximum distance, one is selected randomly. The counterfactual prompt $\tilde{\mathbf{x}}_i^*$ is generated by mixing the original prompted feature from Eq. (11) with the features of the counterfactual partner:

$$\tilde{\mathbf{x}}_i^* = \gamma \cdot \mathbf{x}_i^* + (1 - \gamma) \cdot \mathbf{x}_c, \quad (13)$$

where $\gamma \in [0, 1]$ controls the mix-up ratio, and $\mathbf{x}_c$ denotes the original feature of the counterfactual partner v_c .

As illustrated in Figure 2 (b), this strategy prioritizes mixing between nodes with orthogonal sensitive attributes (e.g., $[0, 1, 1]$ and $[1, 0, 0]$). The Hamming distances are precomputed using Eq. (6) from Section 4.1, requiring no additional parameters or computational overhead during training.

4.3 Multi-Fairness-Aware Adapter Training

In the adapter tuning stage, our objective is to adapt pre-trained GNNs to downstream tasks while ensuring fairness across multiple sensitive attributes. Given an L -layer pre-trained GNN f_θ and input graph $\mathcal{G}$, we apply the prompting function Φ (Sections 4.1 and 4.2) to generate the prompted graph $\mathcal{G}^* = (\mathbf{X}^*, \mathbf{A}^*)$. The GNN computes layer-wise representations as:

$$\mathbf{H}^{(l)} = f_\theta^{(l)}(\mathbf{A}^*, \mathbf{H}^{(l-1)}; \theta^{(l)}), \quad (14)$$

where $\theta^{(l)}$ denotes the parameter of f_θ at the l -th layer, and $\mathbf{H}^{(0)} = \mathbf{X}^*$. The final output embeddings $\mathbf{H}^{(L)} \in \mathbb{R}^{N \times d_L}$ is denoted as $\mathbf{H}$ for simplicity.

4.3.1 Utility Objective. We employ an adapter Ψ to project embeddings to task-specific features $\mathbf{Z} = \Psi(\mathbf{H}^{(L)})$, which are classified by a classifier f_c . The utility objective maximizes prediction accuracy through cross-entropy loss ℓ_{ce} :

$$\mathcal{L}_{\text{util}} = \sum_{v_i \in \mathcal{V}_L} \ell_{ce}(f_c(\mathbf{z}_i), y_i). \quad (15)$$

4.3.2 Contrastive Objective. To enhance fairness, we introduce a contrastive objective that enforces consistency between original and counterfactually perturbed inputs. Let $\tilde{\mathbf{X}}^* = \Phi(\tilde{\mathbf{X}})$ be the prompted counterfactual features from Eq. (13), with corresponding outputs $\tilde{\mathbf{Z}} = \Psi(f_\theta(\tilde{\mathbf{X}}^*, \mathbf{A}^*))$. The contrastive loss is defined as:

$$\mathcal{L}_{\text{cl}} = \sum_{v_i \in \mathcal{V}_L} \ell_{ce}(f_c(\tilde{\mathbf{z}}_i), y_i) + \beta_c \cdot \|\mathbf{z}_i - \tilde{\mathbf{z}}_i\|_2^2, \quad (16)$$

where β_c balances two components: (1) a *counterfactual classification loss* that maintains prediction accuracy on perturbed inputs, and (2) a *consistency loss* that promotes representation invariance under counterfactual perturbations.

4.3.3 Fairness Objective. While adversarial training methods learn invariant representations, they often suffer from instability and require additional discriminators. Inspired by the Information Bottleneck (IB) principle [37], we aim to learn representations that compress sensitive information while preserving predictive power using mutual information (MI). However, directly optimizing MI is computationally challenging. Instead, we employ the Hilbert-Schmidt Independence Criterion (HSIC) [12], a non-parametric kernel-based measure of statistical independence that offers computational efficiency and training stability. For two sets of vectors $\mathbf{U} = [\mathbf{U}_1, \dots, \mathbf{U}_N]$ and $\mathbf{V} = [\mathbf{V}_1, \dots, \mathbf{V}_N]$, the empirical HSIC value is computed as:

$$\text{HSIC}(\mathbf{U}, \mathbf{V}) = \frac{1}{(N-1)^2} \text{tr}(\mathbf{K}_U \mathbf{M} \mathbf{K}_V \mathbf{M}), \quad (17)$$

where $\mathbf{M} = \mathbf{I} - \frac{1}{N} \mathbf{1}\mathbf{1}^\top$ is the centering matrix, $\mathbf{I}$ is the identity matrix, and $\mathbf{1}$ is an all-ones column vector. $\mathbf{K}_U, \mathbf{K}_V$ are kernel matrices of $\mathbf{U}$ and $\mathbf{V}$, respectively. We employ the Gaussian kernel $k(\mathbf{x}, \mathbf{y}) = \exp\left(-\frac{\|\mathbf{x}-\mathbf{y}\|_2^2}{2\sigma^2}\right)$ for its ability to capture nonlinear dependencies across different scales.

The empirical HSIC offers a robust measure of statistical dependence. A smaller $\text{HSIC}(\mathbf{U}, \mathbf{V})$ value indicates stronger statistical independence between $\mathbf{U}$ and $\mathbf{V}$. To ensure the independence of both the model's input and output from sensitive attributes, we formulate the fairness objective as a composite HSIC regularizer:

$$\begin{aligned} \mathcal{L}_{\text{fair}} &= \text{HSIC}(\mathbf{X}^*, \mathbf{S}) + \text{HSIC}(\mathbf{Z}, \mathbf{S}) - \text{HSIC}(\mathbf{Z}, \mathbf{Y}) \\ &= \frac{1}{(N-1)^2} \left[\text{tr}(\mathbf{K}_{\mathbf{X}^*} \mathbf{M} \mathbf{K}_{\mathbf{S}} \mathbf{M}) + \text{tr}(\mathbf{K}_{\mathbf{Z}} \mathbf{M} \mathbf{K}_{\mathbf{S}} \mathbf{M}) - \text{tr}(\mathbf{K}_{\mathbf{Z}} \mathbf{M} \mathbf{K}_{\mathbf{Y}} \mathbf{M}) \right]. \end{aligned} \quad (18)$$

The first two terms in Eq (18) promote independence between prompted features/output representations and sensitive attributes, while the third term acts as a *utility safeguard* by preserving dependence between predictions and labels, preventing over-correction that could harm task performance.

HSIC provides a key advantage by capturing complex dependencies across all sensitive attributes simultaneously. By computing the kernel matrix $\mathbf{K}_{\mathbf{S}}$ over the entire sensitive attribute matrix $\mathbf{S} \in \mathbb{R}^{N \times K}$, we effectively model the joint distribution of sensitive attributes rather than treating them independently.

4.3.4 Overall Objective. The complete optimization objective combines all components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{util}} + \lambda_c \cdot \mathcal{L}_{\text{cl}} + \lambda_f \cdot \mathcal{L}_{\text{fair}}, \quad (19)$$

where λ_c and λ_f balance utility and fairness. During training, only the parameters of the prompt function, the adapter, and the classifier are updated, while the pre-trained GNN remains frozen.

As illustrated in Figure 2 (c), our approach directly addresses *fairness gerrymandering* by ensuring statistical independence between model representations and the sensitive attribute space, preventing bias from shifting to unprotected attribute combinations.

4.4 Theoretical Analysis

In this section, we provide theoretical guarantees for MultiFPT. We first establish the adaptiveness of our approach in achieving equivalent effects to any prompting function. Then, we prove fairness guarantees through Lipschitz continuity analysis.

4.4.1 Adaptiveness Guarantee. Given a pre-trained GNN f_θ and input graph $\mathcal{G} = (\mathbf{A}, \mathbf{X})$, let $\mathcal{G}^* = (\mathbf{A}^*, \mathbf{X}^*)$ be the prompted graph. The candidate spaces for $\mathbf{A}^*$ and $\mathbf{X}^*$ are $\mathcal{A}$ and $\mathcal{X}$, respectively.

THEOREM 4.1. *For any prompted graph $\widehat{\mathcal{G}} = (\widehat{\mathbf{A}} \in \mathcal{A}, \widehat{\mathbf{X}} \in \mathcal{X})$ in the candidate space of $\mathcal{G}^*$, there exists a prompt $\widehat{\mathbf{p}}$ such that:*

$$f_\theta(\mathbf{A}^*, \mathbf{X} + \widehat{\mathbf{p}}) = f_\theta(\widehat{\mathbf{A}}, \widehat{\mathbf{X}}). \quad (20)$$

Theorem 4.1 indicates that our method is highly adaptable. By injecting structural-level and feature-level prompts, the model can achieve effects equivalent to any graph transformation. This allows the PGMs to handle new tasks without parameter updates.

4.4.2 Fairness Guarantees. We establish fairness guarantees through Lipschitz continuity analysis of the composite function $f = f_c \circ \Psi \circ f_\theta \circ \Phi$. The following lemma characterizes the Lipschitz continuity of each component.

LEMMA 4.2 (LIPSCHITZ CONTINUITY OF COMPONENTS). *For any two graphs $\mathcal{G}$ and $\mathcal{G}'$, the following bounds hold:*

- (1) *Prompting function Φ : $\|\Phi(\mathcal{G}) - \Phi(\mathcal{G}')\| \leq L_\Phi \cdot d(\mathcal{G}, \mathcal{G}')$, where $L_\Phi = \max(1, \eta/4)$.*
- (2) *GNN Encoder f_θ : $\|f_\theta(\Phi(\mathcal{G})) - f_\theta(\Phi(\mathcal{G}'))\| \leq L_\theta \cdot \|\Phi(\mathcal{G}) - \Phi(\mathcal{G}')\|$.*
- (3) *Adapter Ψ and Classifier f_c : Both are Lipschitz continuous with constants L_Ψ and L_c , respectively.*

Here, $d(\mathcal{G}, \mathcal{G}')$ denotes a graph distance metric, and L_θ, L_Ψ, L_c are bounded due to spectral normalization in their parameterizations.

The Lipschitz continuity of the prompting function Φ is established as follows: feature prompting $\Phi_X(\mathbf{X}) = \mathbf{X} + \mathbf{P}$ is 1-Lipschitz; structural prompting is based on a sigmoid function, thus is Lipschitz continuous. When optimized under the HSIC regularizer, Φ produces representations invariant to sensitive attributes. The Lipschitz continuity of subsequent components (i.e., f_θ, Ψ, f_c) ensures this invariance propagates to the final predictions. Composing these Lipschitz constants yields the overall model stability.

THEOREM 4.3 (MODEL FAIRNESS). *The composite model $f = f_c \circ \Psi \circ f_\theta \circ \Phi$ is Lipschitz continuous. For any two graphs $\mathcal{G}$ and $\mathcal{G}'$:*

$$\|f(\mathcal{G}) - f(\mathcal{G}')\| \leq L \cdot d(\mathcal{G}, \mathcal{G}'), \quad (21)$$

where the overall Lipschitz constant L is given by $L = L_c \cdot L_\Psi \cdot L_\theta \cdot L_\Phi$.

The proof follows from direct application of the chain rule for Lipschitz functions. Theorem 4.3 guarantees that the invariance of

Table 1: Statistics of five real-world datasets. S.A.: Sensitive Attributes.

Statistics	NBA	Bail	Credit	Pokec-n	Pokec-z
#Nodes	403	18,876	30,000	66,569	67,796
#Edges	16,570	403,977	200,526	729,129	882,765
#Features	95	18	13	266	277
#S.A.	4	5	3	8	8

the representation to sensitive attributes learned by Φ is preserved in the final predictions, thereby promoting multi-attribute fairness.

5 EXPERIMENTS

In this section, we extensively evaluate MultiFPT to address the following research questions (RQs): **RQ1:** How effective is MultiFPT in improving overall fairness? **RQ2:** Can MultiFPT mitigate bias across multiple sensitive attributes and their intersections? **RQ3:** How does each key component contribute to the overall performance? **RQ4:** How sensitive is MultiFPT to key hyperparameters?

5.1 Experimental Setup

We briefly introduce the experimental setup below, with additional details provided in Appendix B.

Datasets. We conduct experiments on five real-world graph datasets: *NBA* [6], *Bail* [21], *Credit* [43], *Pokec-n* [36], and *Pokec-z* [36]. Each dataset contains multiple sensitive attributes (e.g., gender, race, age), which are binarized for fairness evaluation. We summarize the dataset statistics in Table 1.

Baselines. We compare MultiFPT with seven state-of-the-art methods from five categories: (1) *Vanilla GNNs*: **GCN** [24]; (2) *Fairness-aware GNNs*: **FairGNN** [6], **NIFTY** [1], and **FairVGNN** [40]; (3) *Fairness-aware GNNs on multiple attributes*: **FairINV** [49]; (4) *Graph prompting method*: **GPF** [11]; (5) *Fairness-aware prompting method*: **FPrompt** [27].

Implementation Details. We use a 2-layer GCN backbone with hidden dimension 24 for all methods. Results are averaged over 5 independent runs. Models are trained with Adam optimizer at learning rate 0.005 for 1000 epochs, with dropout 0.5 and weight decay 10^{-5} . The adapter consists of a 2-layer MLP with residual connections. For PGMs, we employ GCN with Infomax pre-training.

5.2 Prediction Performance and Fairness (RQ1)

We compare MultiFPT with seven state-of-the-art baselines on the node classification task. The results are summarized in Table 2. From the results, we observe the following:

- MultiFPT consistently achieves superior fairness across all five datasets. Notably, on the Credit dataset, it reduces Δ_{DP} and Δ_{EO} by $\sim 60\%$ compared to baselines, demonstrating strong bias mitigation across multiple sensitive attributes.
- Regarding utility, MultiFPT remains competitive with fairness-aware baselines. On the Bail dataset, its performance is comparable to the top-performing NIFTY and significantly outperforms methods with utility compromise, such as FairVGNN.

These results, further supported by the average improvement (IMP) metric, confirm that our approach effectively enhances multi-attribute fairness with minimal impact on predictive accuracy.

Table 2: Comparison results (mean±standard deviation %) of MultiFPT with baseline methods. In each row, best results are marked in bold, and runner-up results are underlined. IMP (%): the average improvment of MultiFPT over the rest baselines.

Datasets	Metrics	GCN	FairGNN	NIFTY	FairVGNN	FairINV	GPF	FPrompt	MultiFPT	IMP (%)
NBA	ACC (↑)	71.17±0.42	63.47±6.44	65.07±6.00	61.97±2.39	71.08±0.97	71.64±1.08	63.38±2.68	64.32±3.40	-3.41
	AUC (↑)	75.61±1.13	68.35±5.74	71.50±0.16	68.54±0.32	76.43±0.90	<u>75.68±0.82</u>	70.05±1.38	70.33±2.17	-2.54
	$\bar{\Delta}_{DP}$ (↓)	19.54±0.80	<u>15.36±3.43</u>	15.44±3.04	16.08±3.14	20.30±0.83	22.61±2.47	16.85±2.25	13.59±3.77	+23.10
	$\bar{\Delta}_{EO}$ (↓)	13.27±2.40	11.89±3.53	5.95±2.03	8.24±1.25	14.73±1.19	19.64±4.62	11.07±4.24	<u>7.57±3.73</u>	+28.84
Bail	ACC (↑)	<u>90.32±0.55</u>	76.89±22.65	91.75±0.31	82.36±3.88	89.83±1.66	79.78±18.07	78.58±8.90	81.50±5.84	-2.77
	AUC (↑)	90.79±0.80	88.84±2.28	93.45±0.66	88.31±0.51	90.88±2.66	79.26±21.41	<u>91.55±2.27</u>	91.54±1.77	+3.10
	$\bar{\Delta}_{DP}$ (↓)	11.33±0.29	9.50±5.46	10.66±0.37	9.41±0.40	10.96±0.28	12.09±0.88	<u>7.96±1.96</u>	7.83±2.44	+22.46
	$\bar{\Delta}_{EO}$ (↓)	6.83±0.63	6.74±3.90	6.00±0.31	4.82±0.57	6.88±1.02	7.52±1.83	3.54±1.45	<u>4.41±1.52</u>	+22.53
Credit	ACC (↑)	62.97±2.46	60.04±21.87	<u>65.10±6.64</u>	69.57±1.56	64.50±6.35	53.57±4.28	59.33±6.73	60.85±12.50	-1.51
	AUC (↑)	57.03±2.26	57.07±11.17	63.06±1.66	71.36±0.04	<u>65.12±2.16</u>	52.95±4.54	64.74±5.79	55.51±3.01	-9.11
	$\bar{\Delta}_{DP}$ (↓)	4.90±0.84	4.38±4.85	4.77±2.15	9.71±0.82	6.37±2.31	<u>4.34±2.26</u>	7.12±2.51	2.20±1.64	+59.99
	$\bar{\Delta}_{EO}$ (↓)	4.47±0.72	<u>4.08±4.57</u>	4.35±2.40	8.77±0.94	6.14±2.64	4.26±1.75	6.75±2.99	2.01±1.44	+60.99
Pokeyc-n	ACC (↑)	<u>69.72±0.31</u>	65.55±1.60	66.44±1.75	63.80±1.57	69.90±0.44	66.99±0.42	66.51±0.23	61.86±5.07	-7.57
	AUC (↑)	74.19±0.48	69.20±2.33	70.92±1.01	69.92±0.26	<u>74.08±1.35</u>	70.26±0.30	69.55±0.08	67.15±1.65	-5.57
	$\bar{\Delta}_{DP}$ (↓)	18.47±0.33	18.48±3.65	19.25±1.86	14.60±1.50	19.26±0.64	22.24±0.23	18.74±1.41	<u>14.94±5.64</u>	+19.12
	$\bar{\Delta}_{EO}$ (↓)	18.51±0.66	18.80±4.74	19.38±3.08	11.18±1.97	19.44±1.29	23.24±0.29	20.43±2.23	<u>15.89±6.15</u>	+11.09
Pokeyc-z	ACC (↑)	70.14±0.20	67.28±1.21	64.93±4.26	64.24±0.38	70.14±0.35	66.70±0.48	65.61±0.71	64.16±0.31	-4.15
	AUC (↑)	<u>76.96±0.20</u>	71.77±1.47	73.15±1.64	70.95±0.21	77.27±0.09	70.95±1.49	71.71±0.11	69.97±0.45	-4.37
	$\bar{\Delta}_{DP}$ (↓)	16.04±0.56	22.54±1.48	<u>13.36±5.92</u>	13.73±0.10	16.96±1.94	22.31±0.48	14.97±1.99	11.24±1.00	+31.76
	$\bar{\Delta}_{EO}$ (↓)	15.08±1.15	21.94±2.05	11.68±5.26	<u>11.18±0.11</u>	15.87±2.96	21.70±0.32	15.17±1.70	10.77±1.72	+28.92

Table 3: Fairness across individual sensitive attributes on the Pokeyc-z dataset. The fairness-aware baselines are optimized for the sensitive attribute *Region*, while MultiFPT is optimized for all attributes *simultaneously*. Note that FairINV implicitly infers sensitive attributes for optimization. Best results are marked in bold, and runner-up results are underlined.

Method	Optimized S.A.	Region		Gender		Age		Language		HairColor		Smoking		HasChildren		LikeMovie	
		Δ_{DP}	Δ_{EO}	Δ_{DP}	Δ_{EO}	Δ_{DP}	Δ_{EO}	Δ_{DP}	Δ_{EO}	Δ_{DP}	Δ_{EO}	Δ_{DP}	Δ_{EO}	Δ_{DP}	Δ_{EO}	Δ_{DP}	Δ_{EO}
GCN	-	10.20	9.23	3.02	1.59	58.43	44.84	28.40	26.16	17.24	18.98	<u>2.51</u>	9.39	<u>2.14</u>	1.73	6.37	8.74
GPF	-	8.55	6.85	5.02	9.54	63.18	53.60	32.64	30.91	25.75	25.82	19.22	20.99	9.11	12.60	15.02	13.27
FairGNN	Region	8.08	8.00	2.44	5.36	67.09	57.79	34.67	30.24	24.76	25.92	18.20	22.56	6.97	10.43	18.07	15.25
NIFTY	Region	6.32	<u>5.27</u>	3.29	5.43	43.72	34.69	<u>23.64</u>	<u>19.95</u>	<u>13.83</u>	<u>13.24</u>	5.62	6.98	2.75	<u>2.03</u>	7.69	5.86
FairVGNN	Region	<u>5.82</u>	2.00	1.08	3.25	41.53	33.47	26.83	24.31	17.25	17.02	1.85	1.64	8.18	3.51	7.29	<u>4.24</u>
FairINV	Inferred	11.48	10.06	2.10	1.59	60.37	47.22	30.19	27.29	16.68	18.49	5.47	11.39	2.49	2.80	6.86	8.09
FPrompt	Region	5.48	6.87	2.52	7.19	53.50	44.19	23.95	21.45	14.00	15.22	6.95	11.23	4.04	6.45	9.29	8.75
MultiFPT	All	9.26	6.26	<u>1.66</u>	4.79	41.23	<u>33.87</u>	18.67	17.20	10.70	11.62	3.85	<u>6.34</u>	1.72	4.20	2.81	1.90

5.3 Multi-Attribute Fairness (RQ2)

Building on the overall fairness improvements demonstrated in Section 5.2, we further validate MultiFPT's effectiveness in mitigating bias across *individual sensitive attributes* and their *intersections*.

- Table 3 evaluates multi-attribute fairness on Pokeyc-z. While fairness-aware baselines perform well on their optimized target attribute *Region*, they fail to improve fairness on other attributes like *Age* and *Language*. In contrast, MultiFPT, optimized for all attributes jointly, achieves superior fairness across most sensitive attributes, with notable improvements on *Language*, *HairColor*, *HasChildren*, and *LikeMovie*.
- Moreover, Table 4 evaluates fairness on intersectional sensitive attribute groups pairs of *Race* and *Gender* on the Bail dataset. Six

pairs of intersectional groups are evaluated, e.g., "WM-BF" denotes the disparity (Δ_{DP}) between intersectional groups "White Male" and "Black Female". Results show that MultiFPT outperforms baselines in nearly all intersectional groups, with substantial improvements in both mean and maximum fairness disparity. This demonstrates MultiFPT's superiority in mitigating fairness gerrymandering compared to existing methods.

5.4 Ablation Study (RQ3)

We conduct an ablation study to evaluate the contribution of each component in MultiFPT, including the following model variants: **w/ util**: with only utility loss $\mathcal{L}_{util}$; **w/o A***: without structural prompt A*; **w/o X-p**: without learnable prompt P; **w/o X-cf**: without counterfactual prompt X_c ; **w/o X***: without feature prompt X^* ; **w/o G***:

Table 4: Intersectional fairness evaluation on the Bail dataset for the Race-Gender combinations. For example, "WM-BF" denotes the disparity ($\Delta_{DP}\%$) between intersectional groups "White Male" and "Black Female". Similar interpretations apply to other columns.

Methods	Intersectional Group Pairs (Race, Gender)						Mean	Max
	WM-WF	WM-BM	WM-BF	WF-BM	WF-BF	BM-BF		
GCN	11.91	13.56	5.04	25.47	6.87	18.60	13.58	25.47
FairGNN	9.34	10.17	4.43	19.51	4.91	14.60	10.49	19.51
NIFTY	11.86	13.91	4.34	25.76	7.52	18.25	13.61	25.76
FairVGNN	15.86	12.82	11.26	28.68	4.60	24.08	16.22	28.68
FairINV	12.93	13.22	5.43	26.15	7.50	18.65	13.98	26.15
GPF	11.72	10.46	6.50	21.03	5.22	15.81	11.79	21.03
FPrompt	<u>8.26</u>	11.38	3.05	19.64	7.02	<u>12.62</u>	<u>10.33</u>	19.64
MultiFPT	6.30	9.08	0.93	15.38	7.14	8.24	7.85	15.38

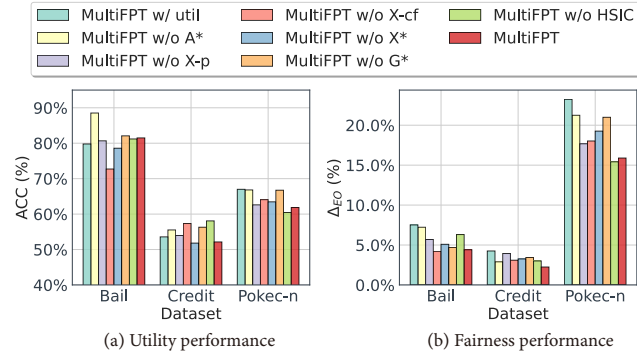


Figure 3: Ablation study of MultiFPT variants on Bail, Credit, and Pokec-n. "w/o" denotes "without".

without both structural and feature prompts; **w/o HSIC**: without HSIC regularizer $\mathcal{L}_{fair}$.

Results in Figure 3 reveals several key findings: *First*, structural prompt removal severely degrades fairness, especially on Bail and Pokec-n, confirming the necessity of topological bias mitigation. *Second*, ablating feature prompt also reduces fairness, with the relative importance of learnable versus counterfactual prompts varying by dataset. *Third*, removing both structural and feature prompts causes substantial fairness degradation across all datasets, highlighting the importance of our prompting framework that addresses both topological and feature-level biases. *Additionally*, removing the HSIC regularizer impairs fairness on Bail and Credit, demonstrating its crucial role in enforcing statistical independence. In summary, each component of MultiFPT makes critical contributions to its overall effectiveness.

5.5 Hyperparameter Sensitivity (RQ4)

We analyze the sensitivity of MultiFPT to two key hyperparameters: the contrastive loss weight λ_c and fairness loss weight λ_f . We vary both values over $\{0.001, 0.01, 0.1, 0.5, 1, 10, 100\}$ on Credit and Pokec-z datasets. Results in Figure 4 show that increasing λ_c significantly improves fairness with a minor utility cost. The model achieves optimal balance when $\lambda_c \in [10, 100]$, maintaining both

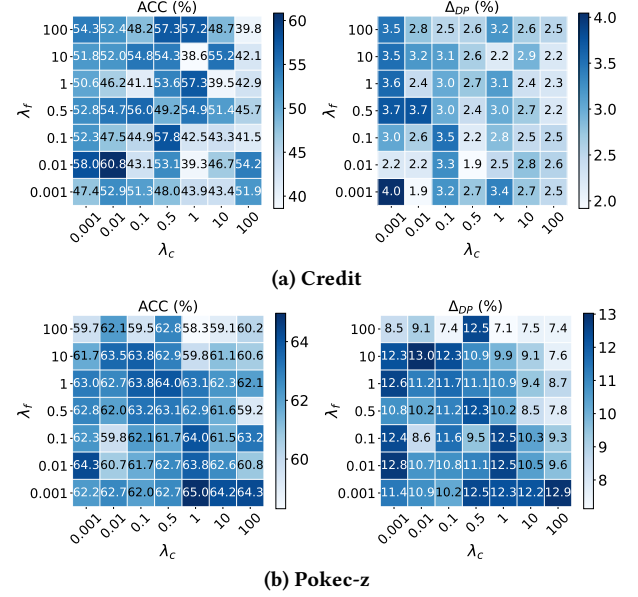


Figure 4: Parameter sensitivity results on Credit and Pokec-z.

high ACC and satisfactory fairness, while values below $\lambda_c = 1$ lead to noticeable declines in both utility and fairness. Similarly, λ_f critically influences fairness. When λ_f is too small (e.g., 10^{-3}), fairness is poorly enforced despite high ACC. Increasing λ_f substantially enhances fairness with only slight utility degradation. Overall, MultiFPT exhibits robustness across a wide hyperparameter range of λ_c and λ_f values. We recommend setting $\lambda_c \in [10, 100]$ and tuning λ_f around 0.5 for optimal performance.

6 CONCLUSION

In this paper, we address the critical challenge of achieving multi-attribute fairness in pre-trained GNNs without model retraining. We propose MultiFPT, a novel graph prompt tuning framework that enhances fairness by injecting lightweight prompts into both graph structure and node features. Our approach adapts PGMs to satisfy multi-attribute fairness constraints through strategic input modifications and HSIC-regularized representation learning, which effectively enforces statistical independence between learned representations and sensitive attributes. Comprehensive experiments on five real-world datasets demonstrate that MultiFPT consistently outperforms state-of-the-art baselines, significantly reducing bias across individual sensitive attributes and their intersections while maintaining competitive predictive utility. The framework provides an efficient and practical solution for deploying fair graph learning systems in real-world scenarios.

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A Algorithm

The algorithm of MultiFPT is as follows:

Algorithm 1 MultiFPT Algorithm

Input: Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, pre-trained GNN f_θ , sensitive attributes $\mathcal{S}$, labels $\mathcal{Y}$

Output: Feature prompt $\mathbf{p}$, adapter Ψ , classifier f_c

```

1: 1. Structure and Feature Prompting
2: for each sampled node pair  $(v_i, v_j)$  do
3:   Compute Hamming distance  $d_{ij}$  as Eq. (6)
4:   Compute edge modification probability  $\mathcal{P}(d_{ij})$  as Eq. (7)
5: end for
6: Generate prompted adjacency matrix  $\mathbf{A}^*$  as Eq. (8)
7: for each node  $v_i \in \mathcal{V}$  do
8:   Compute feature prompt  $\mathbf{p}_i$  as Eq. (9)
9:   Generate prompted feature  $\mathbf{x}_i^*$  as Eq. (11)
10:  Find counterfactual partner  $v_c$  by Eq. (12)
11:  Generate counterfactual feature  $\tilde{\mathbf{x}}_i^*$  as Eq. (13)
12: end for
13: 2. Adapter Training
14: Initialize parameters of  $\mathbf{p}$ ,  $\Psi$ ,  $f_c$ 
15: Freeze pre-trained GNN parameters  $f_\theta$ 
16: for each epoch do
17:   Compute GNN's output embeddings  $\mathbf{H}$  as Eq. (14)
18:   Compute projected embeddings as  $\mathbf{Z} = \Psi(\mathbf{H})$ 
19:   Compute counterfactual embeddings as  $\tilde{\mathbf{Z}} = \Psi(f_\theta(\tilde{\mathbf{X}}^*, \mathbf{A}^*))$ 
20:   Compute utility loss  $\mathcal{L}_{\text{util}}$  by Eq. (15)
21:   Compute contrastive loss  $\mathcal{L}_{\text{cl}}$  by Eq. (16)
22:   Compute fairness loss:  $\mathcal{L}_{\text{fair}}$  by Eq. (17)
23:   Total loss:  $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{util}} + \lambda_c \cdot \mathcal{L}_{\text{cl}} + \lambda_f \cdot \mathcal{L}_{\text{fair}}$ 
24:   Backward pass with  $\mathcal{L}_{\text{total}}$ 
25:   Update parameters of  $\mathbf{p}$ ,  $\Psi$ ,  $f_c$ 
26: end for
27: return  $\mathbf{p}$ ,  $\Psi$ ,  $f_c$ 

```

B Detailed Experimental Setup

B.1 Datasets

We evaluate our method on five real-world graph datasets: NBA [6], Bail [21], Credit [43], Pokec-n, and Pokec-z [36]. Descriptions of these datasets are provided below.

- **NBA** [6] is a social network of professional basketball players, where nodes represent athletes and edges represent follow relationships. We consider four sensitive attributes: *Country*, *Age*, *Height*, and *Points*. Non-binary attributes (e.g., *Age*) are binarized using their average value. The prediction task is to classify whether a player's salary is above or below the median.
- **Bail** [21] contains defendants released on bail in the United States between 1990–2009. Nodes represent defendants, and edges are constructed based on the similarity of their historical criminal records. We use five sensitive attributes: *Race*, *Marital Status*, *Gender*, *Education*, and *Age*. The task is to predict whether a defendant will commit a crime after being released on bail.
- **Credit** [43]: is a credit card user graph where nodes represent users and edges reflect similarity in spending behavior. We adopt three sensitive attributes: *Marital Status*, *Age*, and *Education*. The

goal is to predict whether a user will default on a credit card payment.

- **Pokec-n** [36] is a subgraph of the Pokec social network from the Nitriansky region in Slovakia. Nodes represent users and edges represent friendships. We consider eight sensitive attributes: *Region*, *Gender*, *Age*, *Language*, *HairColor*, *Smoking*, *HasChildren*, and *LikeMovie*. The task is to predict a user's working field.
- **Pokec-z** [36] is another regional subgraph from Pokec, representing users from the Zilinsky region. It uses the same set of sensitive attributes and prediction task as Pokec-n.

Figure 5 shows the cosine correlation among multiple sensitive attributes. The abbreviations of sensitive attributes are shown in Table 5. We observe that most attributes exhibit low correlation, though some attributes in Pokec-n and Pokec-z (e.g., *Language* and *HairColor*) are more closely correlated.

Table 5: Sensitive attribute abbreviations of five datasets.

Dataset	Sensitive Attribute Abbreviations
NBA	CT:Country, AG:Age(>29), HT:Height(<195cm), PO:Points(<5)
Bail	RC:Race, MA:Married, GD:Gender, ED:Education, AG:Age
Credit	MA:Married, AG:Age, ED:Education
Pokec-n	RG:Region, GD:Gender, AG:Age(>25), LA:Language,
Pokec-z	HC:HairColor, SM:Smoking, HC:HasChildren, LM:LikeMovie

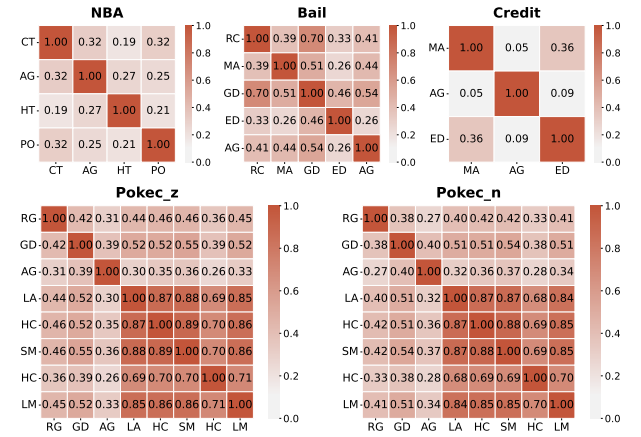


Figure 5: Cosine correlation heatmap across multiple sensitive attributes on five real-world datasets, darker colors suggest a closer correlation.

B.2 Baselines

We compare MultiFPT with the following seven state-of-the-art approaches:

- **GCN** [24]: A widely adopted spectral graph convolutional network for semi-supervised node classification.
- **FairGNN** [6]: A fairness-aware GNN that employs adversarial training and sensitive attribute estimation to enhance fairness with respect to a single sensitive attribute.

Table 6: Fairness performance on Δ_{DP} (%) and Δ_{EO} (%) across sensitive attributes on the NBA dataset. Fairness-aware baselines are optimized for the attribute *Country*, while MultiFPT is optimized for all attributes *simultaneously*.

Method	Optimized S.A.	Country		Age		Height		Points	
		Δ_{DP}	Δ_{EO}	Δ_{DP}	Δ_{EO}	Δ_{DP}	Δ_{EO}	Δ_{DP}	Δ_{EO}
GCN	-	6.88	14.48	32.21	5.04	10.95	8.31	28.13	25.25
GPF	-	8.38	12.99	36.35	10.95	15.17	12.54	30.55	42.08
FairGNN	Country	12.47	12.73	<u>14.39</u>	5.58	10.20	11.72	24.36	17.52
NIFTY	Country	12.00	<u>6.14</u>	26.85	8.32	3.92	3.41	19.00	5.94
FairVGNN	Country	15.90	8.80	23.57	10.73	8.08	<u>7.46</u>	16.76	<u>5.98</u>
FairINV	Inferred	<u>6.55</u>	16.14	31.53	2.29	13.01	10.45	30.12	30.06
FPrompt	Country	11.99	6.97	12.81	4.48	<u>8.05</u>	8.78	34.55	24.04
MultiFPT	All	2.70	4.21	23.08	<u>2.75</u>	10.64	12.96	<u>17.92</u>	10.36

Table 7: Fairness performance on Δ_{DP} (%) and Δ_{EO} (%) across sensitive attributes on the Credit dataset. Fairness-aware baselines are optimized for the attribute *Married*, while MultiFPT is optimized for all attributes *simultaneously*.

Method	Optimized S.A.	Married		Age		Education	
		Δ_{DP}	Δ_{EO}	Δ_{DP}	Δ_{EO}	Δ_{DP}	Δ_{EO}
GCN	-	2.65	3.21	<u>4.70</u>	3.48	7.35	6.72
GPF	-	2.19	2.42	5.70	5.22	5.13	5.12
FairGNN	Married	0.74	0.87	7.90	7.16	4.51	4.19
NIFTY	Married	1.68	2.12	10.70	8.92	<u>1.94</u>	<u>2.02</u>
FairVGNN	Married	3.91	2.84	15.71	15.17	9.51	8.30
FairINV	Inferred	<u>1.17</u>	1.49	13.17	13.02	4.77	3.92
FPrompt	Married	1.54	<u>1.21</u>	10.06	9.15	9.77	9.91
MultiFPT	All	2.16	2.22	3.21	<u>3.52</u>	1.15	1.00

- **NIFTY** [1]: A framework that promotes fairness and robustness via counterfactual augmentation and contrastive learning. It flips sensitive attributes and removes/adds edges to generate augmented views.
- **FairVGNN** [40]: A method that applies adversarial learning and feature masking during message propagation to mitigate sensitive information leakage.
- **FairINV** [49]: A method that adopts invariant learning to mitigate bias on various sensitive attributes from the causal inference perspective.
- **GPF** [11]: A graph prompting framework that introduces learnable feature prompts to adapt pre-trained GNNs to downstream tasks.
- **FPrompt** [27]: A fairness-aware prompting approach that injects both structural and feature prompts into the graph to improve fairness under a single sensitive attribute setting.

B.3 Implementation Details

For MultiFPT, we employ a 2-layer MLP with residual connections and a linear classifier. The dimension reduction factor is set to 4, producing down-sampled embeddings of dimension $\mathbb{R}^{d/4}$. The edge modification ratio is 0.5 for NBA and 0.4 for other datasets. The scaling factor η in Eq. (7) is set to 5, and the feature mix ratio γ in Eq. (13) is 0.8. Each node is assigned 10 learnable prompts, and we use a Gaussian kernel for HSIC computation. Model training is guided by an early stopping criterion that triggers when accuracy degrades by over 2% for 30 consecutive epochs, ensuring stable

utility. For baseline methods, we use a 2-layer GCN backbone with hidden dimension 24. We set the train/val/test ratio of all datasets as 0.5:0.25:0.25. Detailed hyperparameter settings are as follows:

- **FairGNN**: we set the dropout rate as 0.5 and learning rate as 0.01, the loss weights are set as $\alpha = 100, \beta = 1$.
- **NIFTY**: we set the dropout rate as 0.5 and learning rate as 0.01, the loss coefficient is set to 0.4.
- **FairVGNN** and **FairINV**: we set the dropout rate as 0.5 and learning rate as 0.01, and follow the authors' original parameter setting for other hyperparameters.
- **GPF**: we set the dropout rate as 0.5 and learning rate as 0.01, the prompting method is GPF+ with attention-based prompt generation. The number of prompts is set to 10.
- **FPrompt**: we set the dropout rate as 0.5 and learning rate as 0.01. The prompting method is GPF+ with attention-based prompt generation, and the number of learnable prompts per node is set to 10. We follow the authors' original parameter setting for other hyperparameters.

All models are implemented with PyTorch. We build fairness-aware methods based on PyGDebias³. Experiments are conducted on NVIDIA GeForce RTX 4090D with 24GB memory.

C Complementary Experimental Results

C.1 Multi-Attribute Fairness Comparison

We further validate MultiFPT's effectiveness in mitigating bias across *individual sensitive attributes* on NBA, Bail, and Credit datasets. Results are shown in Table 6-7. From the results, we observe that although the fairness-aware baselines perform well in their optimized target attribute, they often fail to improve fairness in other attributes. In contrast, MultiFPT is optimized jointly for all attributes and achieves superior fairness across most sensitive attributes.

C.2 Intersectional Fairness Comparison

We evaluate intersectional fairness on the Credit and Pokec-z datasets in Table 8 and Table 9. MultiFPT outperforms baselines in nearly all intersectional groups, with substantial improvements in both mean and maximum fairness disparity measure (Δ_{DP}). This further demonstrates its capability to mitigate fairness gerrymandering.

C.3 Fairness Comparison of Different PGMs

We evaluate MultiFPT with various PGMs, including GAT [38], GIN [42], and GraphSAGE [13], using the same hyperparameter settings as for GCN. As shown in Table 10, the choice of PGM affects both predictive utility and fairness, with performance varying across datasets. Overall, MultiFPT consistently mitigates bias and improves overall fairness regardless of the underlying PGM architecture.

C.4 Hyperparameter Sensitivity on Bail and Pokec-n

We analyze the sensitivity of MultiFPT to two key hyperparameters: λ_c and λ_f on Bail and Pokec-n datasets in Figure 6. Specifically, we vary the values of λ_c and λ_f as $\{0.001, 0.01, 0.1, 0.5, 1, 10, 100\}$ on Bail and Pokec-n datasets. Results in Figure 6 show that keeping

³<https://github.com/yushundong/PyGDebias>

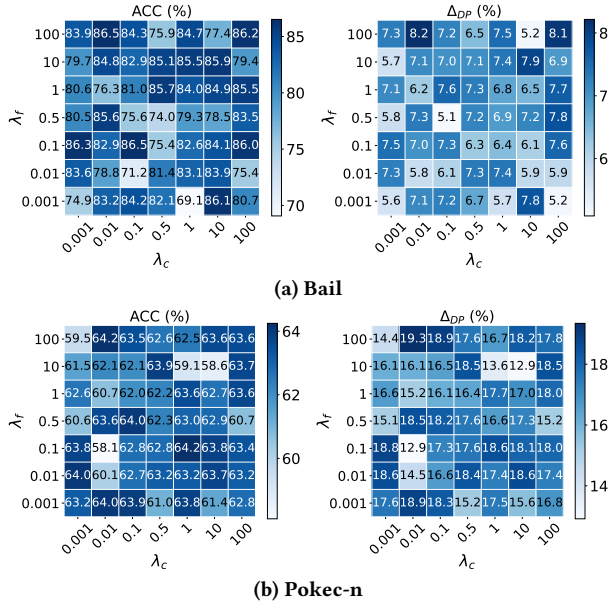


Figure 6: Parameter sensitivity results on Bail and Pokec-n.

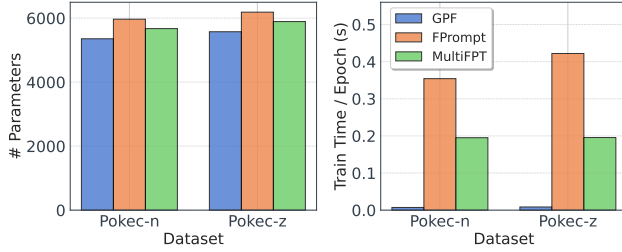


Figure 7: Parameter and training time comparison between MultiFPT and prompting methods (GPF, FPrompt) on the large-scale Pokec-n and Pokec-z datasets.

Table 8: Cross-attribute fairness comparison on the Credit dataset for intersectional group pairs of Age and Education. For example, "YL-OH" denotes the disparity ($\Delta_{DP}\%$) between intersectional groups "Younger, Lower Educated" and "Older, Higher Educated". Similar interpretations apply to other columns.

Methods	Intersectional Group Pairs (Age, Education)						Mean	Max
	YL-YH	YL-OL	YL-OH	YH-OL	YH-OH	OL-OH		
GCN	7.49	4.03	10.43	3.97	7.18	6.91	6.67	10.43
FairGNN	4.62	6.41	6.12	11.01	10.35	0.88	6.57	11.01
NIFTY	2.19	10.15	12.99	9.43	12.27	2.84	8.31	12.99
FairVGNN	9.48	13.52	9.00	23.00	18.48	4.52	13.00	23.00
FairINV	4.90	13.09	19.71	8.19	14.80	6.61	11.22	19.71
GPF	5.11	5.06	4.38	8.64	<u>6.72</u>	4.50	<u>5.73</u>	<u>8.64</u>
FPrompt	10.34	6.26	6.27	16.61	16.61	<u>2.64</u>	9.79	16.61
MultiFPT	1.19	3.09	<u>5.18</u>	2.57	4.48	3.85	3.39	5.18

Table 9: Cross-attribute fairness comparison on the Pokec-z dataset for intersectional group pairs of Gender and Language. For example, "MA-FB" denotes the disparity ($\Delta_{DP}\%$) between intersectional groups "Male, Language A" and "Female, Language B". Similar interpretations apply to other columns.

Methods	Intersectional Group Pairs (Gender, Language)						Mean	Max
	MA-MB	MA-FA	MA-FB	MB-FA	MB-FB	FA-FB		
GCN	34.98	8.68	30.33	26.30	4.65	21.65	21.10	34.98
FairGNN	40.09	12.24	41.03	27.85	1.12	28.79	25.19	41.03
NIFTY	26.25	<u>7.97</u>	28.47	18.47	2.51	20.69	<u>17.40</u>	28.47
FairVGNN	28.46	2.89	<u>27.81</u>	25.78	1.04	25.13	18.52	<u>28.46</u>
FairINV	36.57	9.29	<u>32.87</u>	27.28	3.70	23.58	22.21	<u>36.58</u>
GPF	36.94	12.92	40.69	24.02	3.75	27.77	24.35	40.69
FPrompt	30.95	15.39	31.77	<u>15.56</u>	<u>0.90</u>	<u>16.38</u>	18.49	31.77
MultiFPT	<u>27.51</u>	17.95	27.15	9.56	0.63	9.21	15.33	27.51

Table 10: Overall performance comparison of MultiFPT using different Pre-train GNNs (in %). GSAGE: GraphSAGE.

PGM	Metrics	NBA	Bail	Credit	Pokec-n	Pokec-z
GAT	ACC ($\uparrow$)	53.33	62.24	43.97	61.62	61.25
	AUC ($\uparrow$)	61.85	73.02	52.56	63.98	65.44
	Δ_{SP} ($\downarrow$)	4.47	3.55	3.43	14.21	10.83
	Δ_{EO} ($\downarrow$)	5.83	6.62	3.94	14.37	10.79
GIN	ACC ($\uparrow$)	55.02	70.18	56.10	58.95	59.80
	AUC ($\uparrow$)	57.25	76.68	60.42	61.92	62.55
	Δ_{SP} ($\downarrow$)	12.46	6.53	4.72	12.47	8.55
	Δ_{EO} ($\downarrow$)	12.08	5.67	5.00	13.40	9.65
GSAGE	ACC ($\uparrow$)	53.43	88.41	46.81	61.12	61.40
	AUC ($\uparrow$)	55.21	92.72	53.32	67.69	66.26
	Δ_{SP} ($\downarrow$)	4.88	7.98	2.55	13.35	9.55
	Δ_{EO} ($\downarrow$)	6.05	5.67	2.63	13.39	10.39

λ_c within a certain range is good for improving fairness, i.e., $\lambda_c \in [0.01, 0.5]$ produces relative good fairness results on Bail, and $\lambda_c \in [0.01, 10]$ are good for Pokec-n. For λ_f , too small (e.g., 10^{-3}) or too large values (e.g., 100) both enforce poor fairness results. A medium value around 0.5 is reasonable for both datasets. Overall, MultiFPT exhibits robustness across a wide range of λ_c and λ_f values.

C.5 Efficiency Analysis

We evaluate the efficiency of MultiFPT in terms of parameter count and training speed. We set the number of prompts per node as 10 for all models. As shown in Figure 7, MultiFPT requires fewer parameters and converges faster than the fairness-aware prompting baseline FPrompt. Although GPF is more lightweight, it fails to deliver satisfactory fairness improvements. These results confirm the efficiency of MultiFPT, supporting its practicality in resource-constrained scenarios.